

EDUNEXUS — DAY 1 DEVELOPMENT PROGRESS REPORT

Project: EDUNEXUS — Adaptive Agentic Learning Companion

Event: CEG Agent-a-thon 2026

Development Day: Day 1

Date: 19 September 2026

Development Window: Approximately 9:00 AM – 5:30 PM IST

Repository: ShrineDe06/EDUNEXUS-Agentathon

Overall Day 1 MVP Progress: Approximately 70%

OBJECTIVE

The primary objective for Day 1 was to move EDUNEXUS from its preliminary specification into a working technical foundation with parallel development across the agentic workflow, application backend, and learner-facing frontend.

The team concentrated on four areas:

- validating the supplied Agentic Slice Kit and development environment;
- establishing the EDUNEXUS agent and adaptive mastery-loop architecture;
- implementing and committing the learner-facing frontend;
- preparing the backend, persistence, retrieval, and agent components for end-to-end integration.

The development strategy deliberately retains the supplied agentic infrastructure while extending it with EDUNEXUS-specific learning behaviour.

ENVIRONMENT AND STARTER-KIT VALIDATION

The provided Agentic Slice Kit was successfully configured before application-specific development began.

The environment doctor confirmed:

- .env configuration loaded successfully;
- OpenRouter API key validated;
- primary model reachable;
- fallback model reachable;
- escalation model reachable;
- sqlite-vec v0.1.9 successfully loaded;
- API budget available;

- local environment ready for agent development.

The remaining cloudflared and embedding-model notices were identified as non-blocking warnings rather than application failures.

Status: Completed

Progress: 100%

BASELINE TEST VERIFICATION

The complete existing test suite was executed before modifying the starter architecture.

Baseline result:

74 tests passed

This provides a clean reference point against which subsequent EDUNEXUS-specific changes can be validated.

The team can therefore distinguish failures introduced during EDUNEXUS development from behaviour already present in the supplied starter kit.

Status: Completed

Progress: 100%

REPOSITORY AND GIT WORKFLOW

The supplied repository has been retained as the primary development repository rather than replacing the starter architecture with an independent project.

A pre-development baseline was marked using:

hackathon-start

Feature-oriented development has subsequently been used to separate responsibilities and reduce integration conflicts.

The repository has now progressed beyond its initial documentation-only state and contains application-specific development areas including agent, API, web, corpus, prompt, testing, and domain components.

This structure allows frontend, backend, agent, retrieval, and verification work to proceed in parallel while preserving the common starter infrastructure.

Status: Active and operational

Progress: 95%

AGENTIC ARCHITECTURE ANALYSIS

The team studied the supplied architecture before extending it.

The reusable infrastructure includes:

COMPONENT	PRIMARY RESPONSIBILITY
slice/records.py	Run-state and record definitions
slice/store.py	Durable append-only persistence
slice/config.py	Environment configuration
slice/budget.py	Token and attempt limits
slice/llm.py	Centralised model invocation
slice/retrieve.py	Chunking, embeddings and retrieval
slice/callback.py	Human suspension and resume
slice/runner.py	Deterministic state-machine execution
demo/	Domain-specific agent behaviour
web/	Human-facing interaction layer

A major Day 1 decision was to **extend this architecture rather than rebuild equivalent infrastructure independently**.

Model calls are intended to provide judgement within individual steps, while deterministic application code controls sequencing and state transitions.

Status: Completed

Progress: 100%

ADAPTIVE MASTERY LOOP

The central EDUNEXUS agentic workflow has been defined around mastery rather than one-time question answering.

The core learning cycle is:

Diagnostic → Analyse → Identify Weakness → Remediate → Verify → Mastery

The critical agentic backward transition is:

Verification Failure → Remediation → Verification

This allows EDUNEXUS to react when the learner has not actually corrected a misconception rather than prematurely marking the topic as mastered.

The representative validation scenario currently focuses on:

Topic: Python Lists

Subconcept: Slicing

Detected misconception: Stop index treated as inclusive

Expected recovery: Targeted remediation followed by verification and, where necessary, another remediation cycle.

The workflow is also designed with bounded remediation attempts so that unsuccessful verification cannot create an uncontrolled loop.

Status: Core architecture established; implementation/integration continuing

Progress: 75%

AGENT RESPONSIBILITIES

The learning workflow has been decomposed into distinct responsibilities.

Diagnostic responsibility: interprets learner responses and identifies demonstrated weaknesses.

Assessment responsibility: determines whether available evidence indicates mastery or a specific misconception.

Remediation responsibility: provides targeted teaching for the identified weak concept.

Verification responsibility: checks whether the learner has corrected the misconception.

Progress/Memory responsibility: maintains persistent evidence of concepts encountered, misconceptions detected, remediation attempts, and eventual mastery.

This decomposition prevents EDUNEXUS from becoming a single unrestricted LLM prompt and makes individual stages observable and testable.

Status: Architecture established; implementation active

Progress: 70%

FRONTEND IMPLEMENTATION — DAY 1 MILESTONE

A significant Day 1 milestone was the progression of the learner-facing interface from the planning/prototype stage into a **repository-committed frontend implementation**.

The frontend work establishes the presentation layer through which the learner will ultimately interact with the EDUNEXUS learning workflow.

The interface is being developed separately from the internal agent-control logic so that presentation behaviour does not directly contain model orchestration or mastery decisions.

The intended interaction boundary remains:

Learner → Frontend → Backend/API → EDUNEXUS Agent Workflow

This separation is important because the frontend can evolve independently while the mastery state machine remains deterministic and testable.

With the frontend implementation now committed, the next priority is not additional isolated UI development but **connecting the learner experience to live backend and agent state**.

Status: Core frontend implementation committed; integration continuing

Progress: 80%

BACKEND AND API LAYER

The repository now provides a dedicated application/API area for connecting the user-facing application with the EDUNEXUS agent workflow.

The backend is intended to provide a controlled boundary between interface actions and agent execution rather than placing model calls directly inside the frontend.

The target architecture is:

Frontend → API → Agent Flow → Store / Retrieval / LLM

The primary remaining Day 2 responsibility is to complete and verify the integration between these layers.

Status: In development / integration

Progress: 65%

HUMAN-IN-THE-LOOP LEARNING

The learner remains an active participant in the mastery workflow.

When a weakness is identified, the intended control flow provides a learner decision:

Weakness Detected → Learner Decision → Revise / Skip

A learner choosing **Revise** proceeds into targeted remediation.

A learner choosing **Skip** should have the weakness deferred rather than falsely recorded as mastered.

The supplied callback infrastructure has been identified as the appropriate mechanism for suspending and later resuming these learning runs.

Status: Designed; end-to-end integration pending

Progress: 45%

DOCUMENT-GROUNDED LEARNING AND RETRIEVAL

EDUNEXUS is designed to support learning from uploaded educational material in addition to general curriculum topics.

The supplied retrieval infrastructure provides:

- document chunking;
- embeddings;
- vector search;
- SQLite-backed retrieval.

sqlite-vec has already been successfully validated in the development environment.

The remaining work is to connect document ingestion and retrieval to the actual diagnostic, learning, revision, and assessment experiences.

This is important because document-based interactions must remain grounded in the learner's supplied material rather than producing unsupported content.

Status: Infrastructure validated; application integration continuing

Progress: 40%

TESTING AND RELIABILITY

The project began Day 1 with a verified baseline of:

74 / 74 tests passing

The EDUNEXUS testing strategy separates deterministic workflow testing from live model testing.

Core state transitions should therefore be testable without consuming model credits.

Priority behaviours include:

- weakness detection;
- remediation initiation;
- failed verification;
- backward transition into another remediation cycle;
- successful verification;
- mastery recording;
- remediation limits;
- persistence across steps;
- learner callback behaviour.

This approach allows model variability and application-control defects to be diagnosed independently.

Status: Baseline verified; EDUNEXUS-specific verification continuing

Progress: 60%

API BUDGET AND MODEL STRATEGY

The model environment was validated successfully with the allocated API budget available at the beginning of development.

The team is intentionally avoiding unnecessary live-model calls during deterministic workflow development.

The primary model is used for routine learning operations, while the higher-cost escalation model is reserved for situations where increased model capability is demonstrably necessary.

This reduces unnecessary cost and allows most workflow behaviour to be tested without network dependency.

Status: Completed and actively followed

Progress: 100%

DAY 1 ENGINEERING DECISIONS

The following decisions were established during Day 1:

1. Retain and extend the supplied Agentic Slice Kit.
2. Preserve generic infrastructure instead of rebuilding the state machine.
3. Separate learner-facing UI from agent orchestration.
4. Keep model judgement separate from deterministic sequencing.
5. Implement an explicit backward transition following failed verification.
6. Persist evidence of learner progress rather than relying solely on conversational context.
7. Bound remediation attempts to prevent uncontrolled loops.
8. Preserve learner agency through Revise/Skip decisions.
9. Test deterministic workflow behaviour before relying on live model calls.
10. Develop frontend, backend, agent, and retrieval components as parallel workstreams before final integration.
11. Maintain meaningful Git history so that development, testing, and subsequent fixes remain traceable.

DAY 2 PRIORITY

Day 2 will primarily be an **integration and verification day** rather than an architecture-design day.

The highest-priority tasks are:

- complete the adaptive mastery loop;
- connect frontend interactions to backend endpoints;
- connect backend endpoints to the agent workflow;
- complete learner Revise/Skip callback behaviour;
- connect uploaded-document retrieval to learning and assessment;
- persist misconceptions and mastery history;

- validate the complete LEARN / REVISE / TEST journey;
- conduct real-user walkthroughs;
- perform an adversarial stress test;
- record feedback and implement at least one visible feedback-driven fix;
- prepare the final demonstration and technical explanation.

DAY 1 OVERALL ASSESSMENT

Day 1 successfully moved EDUNEXUS beyond specification and environment setup into **parallel implementation**.

The strongest completed foundations are:

- validated development environment;
- clean automated-test baseline;
- reusable agentic architecture understood;
- Git development workflow established;
- adaptive mastery loop designed;
- frontend implementation committed;
- backend/API and agent workstreams established;
- retrieval infrastructure validated.

The principal remaining risk is now **integration**, not initial architecture.

The next milestone is to demonstrate that a learner action in the committed frontend can travel through the backend into the persistent agent workflow, produce a meaningful adaptive response, and return to the interface while preserving learner state.

Estimated Day 1 MVP Completion: 70%

This percentage represents completion of the **planned hackathon MVP**, not completion of the broader long-term EDUNEXUS product.

FINAL DAY 1 PROGRESS TABLE

Workstream	Day 1 Deliverable	Current Status	Completion
Environment setup	API, models, .env, SQLite vector extension validated	Completed	100%
Starter validation	74 baseline tests successfully executed	Completed	100%
Git/repository workflow	Baseline and parallel development structure established	Operational	95%
Architecture analysis	Starter infrastructure and domain boundaries understood	Completed	100%
Adaptive mastery loop	Diagnostic → remediation → verification → backward recovery designed	Active	75%
Agent layer	Diagnostic, assessment, remediation, verification and progress responsibilities established	Active	70%
Frontend	Learner-facing frontend implementation committed to repository	Implemented / Integration Pending	80%
Backend/API	Interface between frontend and agent workflow under development	Active	65%
Human-in-the-loop	Revise/Skip learner decision designed	Integration Pending	45%
Document/RAG pipeline	Vector infrastructure operational; application grounding underway	Active	40%
Automated testing	Baseline green; domain-specific tests continuing	Active	60%
API/model strategy	Budget-aware model strategy established	Completed	100%
End-to-end integration	Frontend/backend/agent connection still being consolidated	Active	40%
Real-user verification	Walkthrough, feedback and stress-test evidence remain to be completed	Pending	10%
Overall Hackathon MVP	Core foundations and frontend established; integration and verification are next	On Track	~70%


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Mastery progress 72%

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Complete

Slicing patterns

In progress

Comprehensions

Up next

12 concepts

Mastered this week



ENG IN 94% 5:51 PM 9/19/2026


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Each mode shares the same learner profile, so every interaction makes the next one more useful.



BUILD UNDERSTANDING
Learn
 Explore new concepts with an AI tutor grounded in your course material and your learning history.

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STRENGTHEN RECALL
Revise
 Return to weak areas with focused explanations shaped by your previous attempts and misconceptions.

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 Run a focused diagnostic and receive clear, actionable feedback on exactly what to improve next.

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Built around your learning journey
 Five specialized agents coordinate retrieval, diagnosis, remediation, and verification through one persistent learner profile.

SAFE BY DESIGN


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Learn mode — AI tutor
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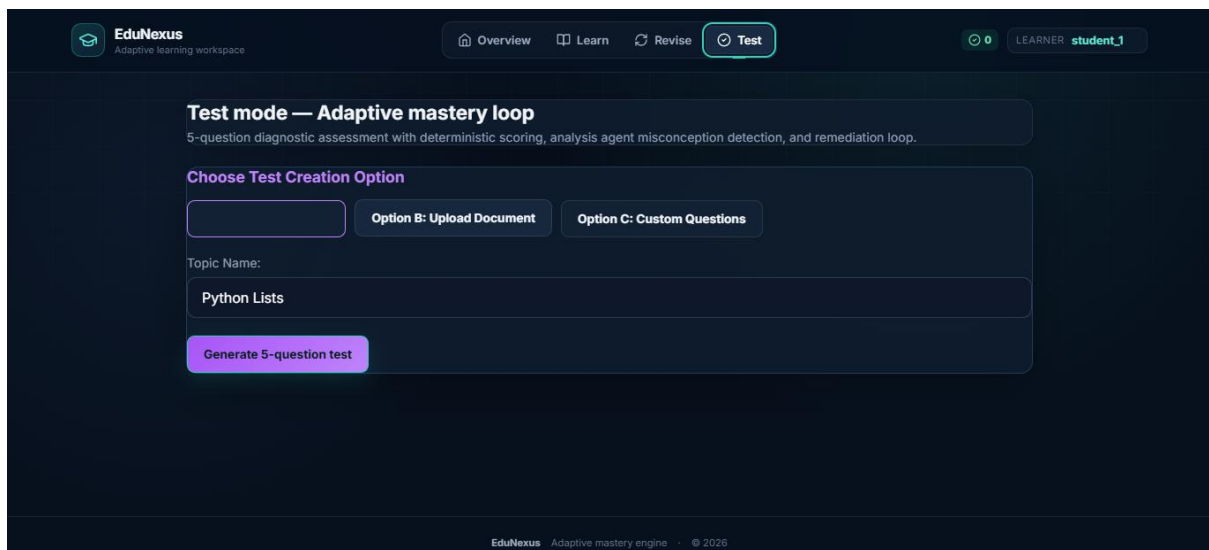
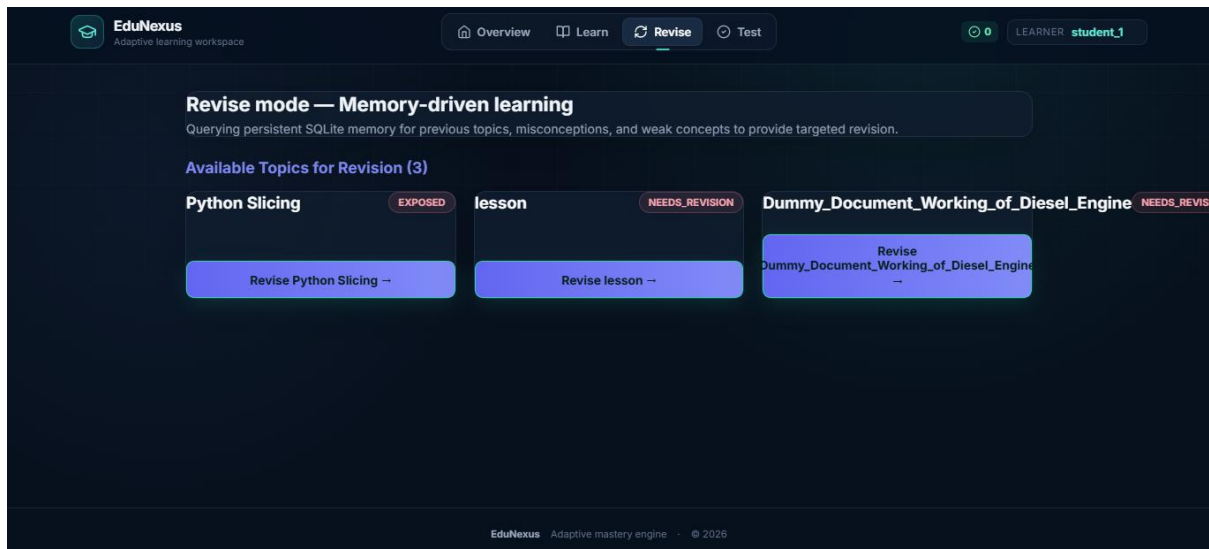
CURRENT TOPIC
 IC engines

ACTIVE DOCUMENT
 None (General Knowledge)

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Day 1 Closing Statement

At the end of Day 1, EDUNEXUS has progressed from an architectural specification into a structured implementation with a validated agentic foundation and a committed learner-facing frontend.

The project now has the principal components required for the MVP represented within the repository. Day 2 will focus on connecting these components into a reliable end-to-end adaptive learning experience and validating that experience with real users and observable state transitions.

Day 1 Status: ON TRACK — approximately 70% of the planned hackathon MVP completed.

END OF DOCUMENT

19 September 2026